

Study Resources: Policy Optimization & Causal Inference

Curated to accompany the 4-Week Learning Plan



Phase 1: Foundations (The "Why")

Must-Read Book (Free & Python-based)

- **Causal Inference for the Brave and True** by Matheus Facure.
 - *Focus*: Read **Part I (The Brave and True)**. It covers potential outcomes, confounding, and the basic math with humor and Python code.
 - *Specific Chapters*: "The Brave and True", "The Unreasonable Effectiveness of Linear Regression", "Propensity Score".

Videos

- **Crash Course in Causal Inference** (YouTube - Microsoft Research).
 - *Why*: A solid high-level overview if you prefer video to text.

Concepts to Google

- "Simpson's Paradox python example"
 - "Potential Outcomes Framework explained"
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Phase 2: Propensity & Weighting (IPW)

Tutorials & Articles

- **A Hands-on Guide to Inverse Propensity Weighting in Python** (Towards Data Science).
 - *Why*: Step-by-step implementation using `pandas` and `sklearn`, exactly relevant to the Kaggle notebook approach.

- **Causallib: IPW Implementation.**
 - *Why:* IBM's library is great for standard IPW. Look at their "examples" folder in the repo.

Libraries to Explore

- `sklearn.linear_model.LogisticRegression` (The workhorse for estimating scores).
 - `causalib` (Dedicated library).
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Phase 3: Doubly Robust Estimation

Readings

- **"Doubly Robust Estimation" in Causal Inference for the Brave and True.**
 - *Why:* Best explanation of *why* combining two "wrong" models can render a "right" result.
- **EconML Documentation: Doubly Robust Learning.**
 - *Why:* Microsoft's EconML is the industry standard. Even if you code from scratch, read their docs for the theory.

Code Examples

- **Simple Doubly Robust Implementation.**
 - *Why:* A minimal, copy-pasteable example to understand the mechanics.
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Phase 4: CATE & Policy Optimization

The "Holy Grail" Libraries

- **CausalML (Uber).**
 - *Focus:* Look at their **Uplift Modeling** examples. They are the leaders in this space (Marketing/Promotions).
- **EconML (Microsoft).**
 - *Focus:* **CATE** estimation using `CausalForest` or `DML` (Double Machine Learning).

Readings on "Meta-Learners"

- [Meta-Learners for Estimating Heterogeneous Treatment Effects](#).
 - *Concepts*: Learn the difference between **S-Learner** (Single model), **T-Learner** (Two models), and **X-Learner** (Cross-over). The Kaggle notebook uses a T-Learner approach.

Policy Optimization Specifics

- [Uplift Modeling: A Quick Introduction](#).
 - *Key Chart*: **Qini Curve** (The "ROC Curve" of causal inference). Learn how to interpret it.
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Recommended Setup for your Practice

Since you are on Mac, I recommend setting up a fresh environment:

```
# Create a dedicated environment
python3 -m venv venv_causal
source venv_causal/bin/activate

# Install the "Big 3" + Standard DS stack
pip install causalm1 econml causallib pandas numpy scikit-learn matplotlib seaborn
```